

Assessing Financial Literacy Engagement and Learning Through Data Visualizations and AI

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Abstract

We developed a financial literacy web app for young adults to explore whether an AI coach improved engagement and learning. While financial literacy is especially important for young adults making major future decisions, our results showed the AI coach did not improve outcomes. Average engagement was 14.9/30 with the coach versus 17.9/30 without, and self-assessment improved by 0.2 compared to 0.8. The AI coach was used mainly for comprehension check support and clarifying information.

Acknowledgements

We sincerely appreciate the guidance and support of our advisors, Professor Stacy Shaw and Professor Lane Harrison, throughout this project. Their expertise, feedback, and encouragement have been impactful in shaping our work, and we are grateful for their time and dedication.

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1 Introduction

In today's modern era of technology, people have an increasing amount of control over their finances and an increased responsibility for financial decision making. The growth of the financial technology (fin-tech) industry in the way of increased payment options and more complex financial products, has increased the demand for financial literacy in the current economic landscape [1]. Furthermore, government-funded retirement benefit accounts such as Social Security are quickly depleting, shifting the general public's need to focus on building their own retirement savings accounts. Finally, people are increasingly responsible for their financial independence through their decision-making [2].

Research has shown that poor financial literacy not only harms individuals but also harms society as a whole. At the individual level, it leads to debt accumulation, foreclosures, and high interest rates. On a governmental level, it can lead to drug addictions, child neglect, and increased use of welfare programs [3]. On the contrary, a society with higher financial literacy has significant benefits. Research has shown that financially literate individuals are more likely to plan for retirement, prepare for economic shocks, and keep debt at a manageable level [4]. This can reduce the burden and reliance on government welfare programs. Some researchers argue that investment in financial literacy should be viewed as an investment in human capital [3] [5].

Despite its glaring importance, financial literacy remains low among people of all ages in most countries around the world [5]. A meta-analysis of the financial literacy space revealed that financial decisions made between the ages of 18 and 24 are likely to impact a person's entire life [6]. However, research shows that financial literacy is specifically low for high school and college students [4] [7] [8]. While there remains some variation of financial literacy levels across various demographics [4] [8], the need to improve financial literacy among all is imminent.

Researchers of the topic have varying definitions of financial literacy. Aisa Amagir (and colleagues) defined it as, "...knowing what to do, the skills to be able to perform these actions, and the inclination to carry them out." Lusardi and Mitchell defined it as "people's ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt, and pensions." Rodriguez-Correa (and colleagues) defined financial literacy as "the ability of people to make adequate decisions regarding their financial life based on judgments created from training in topics about the nature and circulation of money, credit, savings for education, retirement, accidents, pensions, among others." While there is no widely accepted definition of financial literacy, there are key themes that are found in most definitions,

such as understanding basic financial concepts (i.e., savings, debt, wealth accumulation, etc.), making smart decisions based on life circumstances (i.e., current budget, inflationary period, etc.), and the confidence to make those decisions. For the purposes of our research paper, we will define financial literacy as: the ability to understand essential financial concepts and the confidence to apply that understanding to make sound financial decisions within a given economic context.

Due to its profound importance and evident need, researchers have begun investigating ways and keys to teaching financial literacy. Some research has identified the need to build up the confidence of people to make financial decisions [3]. Others have highlighted the importance of teaching financial literacy in late adolescents and with parental guidance who may have their own financial knowledge based on experience and that financial literacy should be highly tailored to an individual's situations [7]. Additionally, researchers have highlighted that simply exposing individuals to financial literacy courses could impact the financial literacy of those around them [4]. Finally, there has been an emphasis that financial literacy could be best taught through hands-on experiences and through the use of visual tools [2] [3].

This introduces the perfect opportunity to use data visualization and web applications to help drive financial literacy improvement. Data visualization has shown tremendous uses for breaking down complex data into understandable charts where insights can be made across a variety of disciplines [9]. Similarly, web applications and online learning has made significant strides in recent years especially during the COVID-19 pandemic and have shown the ability to not only improve learning outcomes for all ages, but motivation to learn [10] [11]. In this project we will leverage data visualization, Artificial Intelligence (AI), and web application teaching to boost financial literacy among college students.

2 Background

In this project, we draw from fields of financial literacy, technology acceptance, educational web applications and artificial intelligence (AI) chatbots. In the financial literacy literature we seek to examine key financial topics and literacy gaps. Additionally, we examine the technology acceptance framework introduced by Davis in 1989 which outlines key factors in user adoption of new technologies [12]. Finally, we review education web applications and their utility as well as the growing field of AI chatbots in educational experiences and environments. This section reviews these key topics and fields in our project.

2.1 Financial literacy

2.1.1 Financial Literacy in the AI Context

In this project we are creating a learning experience to boost the financial literacy of college aged students and early graduates. This group has become specifically vulnerable and has a lower financial literacy [4] [7] [8]. In addition, the expansion of financial technologies has made it easier than ever to spend money and make irresponsible financial decisions [1]. This group also faces the highest debt in history [13][14] and high unemployment rates [15]. For these reasons, delivering effective financial learning experiences to this group is necessary.

A recent project that focused on creating an AI agent to help young adults make responsible financial decisions found that young adults often struggle the most with budgeting and saving strategies [16]. Other research has highlighted gaps in investing and general financial knowledge [7].

Implementing AI in these delicate topics is of high importance. With poor implementation, it could lead to distrust in these AI solutions. Some research suggests that people often trust human drivers over automated driving solutions [17], suggesting poor AI implementation on sensitive or important issues could be problematic. Recent research found that designing AI with more emotion and self-blaming strategies could lead to increased user trust and engagement [18]. In order to ensure reliable results, it's important to implement the AI in our web application in a way people will trust and engage with it.

2.2 Technology Acceptance

Understanding the motivations and drivers why users accept technology and web applications, more specifically educational applications are a key area of research for our project. There's a wide array

of research into technology acceptance, which is getting someone to accept and use your technology or application. Research has identified three key reasons users adopt new technologies: perceived ease of use, perceived usefulness and perceived enjoyment [12]. There has also been significant research into educational application adoption and the keys for this [10] [11] [19] [20] [21]. Research of students aged 20-22 found that use of mobile learning games was effected by this technology acceptance framework of: perceived ease of use, enjoyment and usefulness [11]. This research and frameworks they set up were key factors when designing our financial literacy application and the individual modules. Learning applications and games without careful design has been found to have negative learning outcomes [11].

2.2.1 Perceived Ease of Use

Perceived ease of use is how easy the technology feels to use. Can users easily navigate the platform? Are instructions and activities clear to the user? Is the user guided through to the module without confusion? These are all questions we considered when designing our web application. Research into a mobile learning application designed to help first year students transition into university found that the system quality of the application (ease of use and ease to find things) was key in improving user satisfaction. They also discovered a link between user satisfaction and learning effectiveness, linking ease of use to learning outcomes [21]. Some design features we added to create a high perceived ease of use were making buttons clear and well differentiated, auto-scrolling to parts of the page we want the user and adding hover and click animations when the user is on a button.

2.2.2 Perceived Usefulness

Perceived usefulness is how useful the user thinks the technology is going to be. Research has proven that the quality of information a technology or application provides positively contributes to a user's satisfaction of the application [21]. Additionally, research indicates that adults are likely to adopt a technology and encourage their child to use it if there is a perceived learning benefit [19][10]. To increase the perceived usefulness of our web application we include interactive downloadable data visualizations so users could use the data visualization later for their own use. We also included animated/simulated data visualizations to provide concrete real-world examples to users to increase the perceived usefulness.

2.2.3 Perceived Enjoyment

Perceived enjoyment is how much enjoyment the user is having using the technology. A key way to increase enjoyment is to “gamify” it. That is, to make the learning experience feel like a game where the user gets challenges, tasks and sees their progress. Research found that this gamification can foster a user’s perceived enjoyment [10]. However, it’s important that this gamification is done to still keep the focus on educational outcomes [11]. One way we sought gamification was to add progress bars on top of each lesson and to add achievements after completing a lesson. Another way in which researchers have found to increase perceived enjoyment is through emojis and icons that can make the application feel more like a fun experience and less like a strict learning experience [22][23]. For this, we created an application mascot (a cartoon piggy bank) who pops up and “teaches” users (text information pops up in a speech bubble next to the mascot).

2.2.4 Technology Acceptance in the Context of Education Applications

The technology acceptance framework provides a strong baseline for how to better engage users and encourage them to adopt our web application. There are other key design elements that have been identified specifically in the context of successful educational applications. A research paper studying the motivation of students aged 15-25 to adopt educational applications identified the following traits for successful educational applications: academic assistance, convenience, entertainment, social influence, novelty, engagement, and activity. While some of these can fall within the existing framework, some notable new ones are in social influence, if many people adopt a technology and it is perceived as popular more users may adopt it, and novelty, if users find the application new and exciting they’re more likely to adopt it [19]. In this area we sought to design unique interactive data visualizations that would be novel to what users may have seen. Another study identified the users’ perceived control in the application as another trait of technology acceptance. That is, if a user feels like they control their experience in the application [10]. To cater to this, we build data visualizations where users could input their own data or change the visualization to give them more control over what they are seeing. Additionally, we added open ended response areas to give users more freedom in their responses.

2.3 Educational Web Applications

Web applications to learn a variety of topics have gained significant popularity with the rise of the internet and as technology improves. Education web applications allow wider access to learning environments

and allow users to engage and learn material asynchronously as they want without the requirement of a physical person teaching them. These applications have proved successful across a variety of domains and for a variety of users [10] [19] [21].

Design consideration of educational web applications depends highly on the target audience and domain of the web application. For example, research suggests that educational web applications for younger audiences should feature game-like elements and fun emojis [10], while applications geared more toward university and high school students should focus more on functionality and quality of information provided [19] [21].

The design and information provided is crucial for the effectiveness of an educational web application and thus, is important for the present study. Research indicates that well designed educational applications can boost educational outcomes over traditional classroom learning but only if well designed [20]. On the contrary, poorly designed educational applications can negatively affect learning outcomes and user engagement [11].

In order to ensure the success of our study the design and information provided in our financial education application was carefully considered and designed.

2.4 Chatbots' Impact on Learning

In this study we seek to compare the learning outcomes and user engagement of a chatbot learning experience compared to a self-paced web application learning experience specifically within the context of financial literacy of college students. A chatbot is defined as “a computer program that uses artificial intelligence (AI) to conduct a conversation via auditory or textual methods and interacts with humans in their natural languages. These interactions usually occur through websites, messaging applications, or mobile apps, where the bot is capable of simulating and maintaining human-like conversations and performing different tasks” [24]. Chatbots have been around since 1966 through ELIZA [25] but have gained more popularity in recent years through chatbots like ChatGPT, Gemini, Siri and Alexa. This popularity has led to an increased focus on how they can impact learning.

Chatbots have characteristics that could make them valuable in the learning experience. They have become highly adaptable to user preference and can deliver information to users based on the user's preferences such as providing videos for certain learners or explaining topics in useful ways for the user (i.e. explaining something in sports terms for someone knowledgeable in sports) [26] [27] [28] [29]. Additionally, the conversational aspect of chatbots allows them to receive feedback and adjust in real time [30]. Chatbots

also provide information on demand and respond in close to real time, allowing users to get educational help and learn at their convenience and pace [31] [32].

Chatbots can also be effective for data collection. Recent research suggests that chatbots lead users and respondents to be more open and detailed about their experiences and life [33] [34]. Many users cited chatbots as non-judgemental and understanding, encouraging them to share more information as opposed to personal interviews [33]. This can be important for companies and researchers to be able to extract more data and information, to better understand their users.

Despite this, results on their impact on learning outcomes and user engagement have been mixed. A recent study of college students in a Computer in Education course found that while students who used chatbots in their learning experience did have higher learning outcomes and engagement, the findings were not significant [35]. Other research has shown that chatbots are promising for improving learning outcomes and user engagement [36] [37], while others have shown that chatbots don't improve learning outcomes and user engagement [38].

Chatbots also present some dangers especially in use for learning. Chatbots may be trained on biased data and the use of the chatbots may perpetuate some of these biases. Additionally, there remains inaccuracies of chatbot responses and issues of hallucination of facts that don't exist [39]. It is imperative that these issues are kept in mind when using chatbots for educational purposes and to take measures to avoid them [26].

3 Methods

3.1 App Design and Development

3.1.1 Design

An initial mock up of the application design was created using Figma, a design software platform. Upon group agreement on the Figma design it was implemented into our web application. The design includes several features to go along with the theme of financial literacy and money. The web application is named “MakeNCents” as a play on words on how our application aims to improve people’s financial literacy. A friendly piggy bank mascot, named Penny, is also heavily featured throughout the application to keep users engaged and to go along with the theme. The site also includes various shades of green to symbolize money to further re-enforce the theme.

Our initial design implemented a self-paced Duolingo style learning process where users are presented information and can understand it at their own pace. From there, users can click continue buttons to progress through the lessons. This ”click and continue” style of learning has proven successful in many learning applications. It also gives users the freedom to learn as they please, which is a key component to technology acceptance [10]. Implementing our site this way also allows us to see which pages and information are users are most engaged with.

3.1.2 Figure of Initial Mock-up

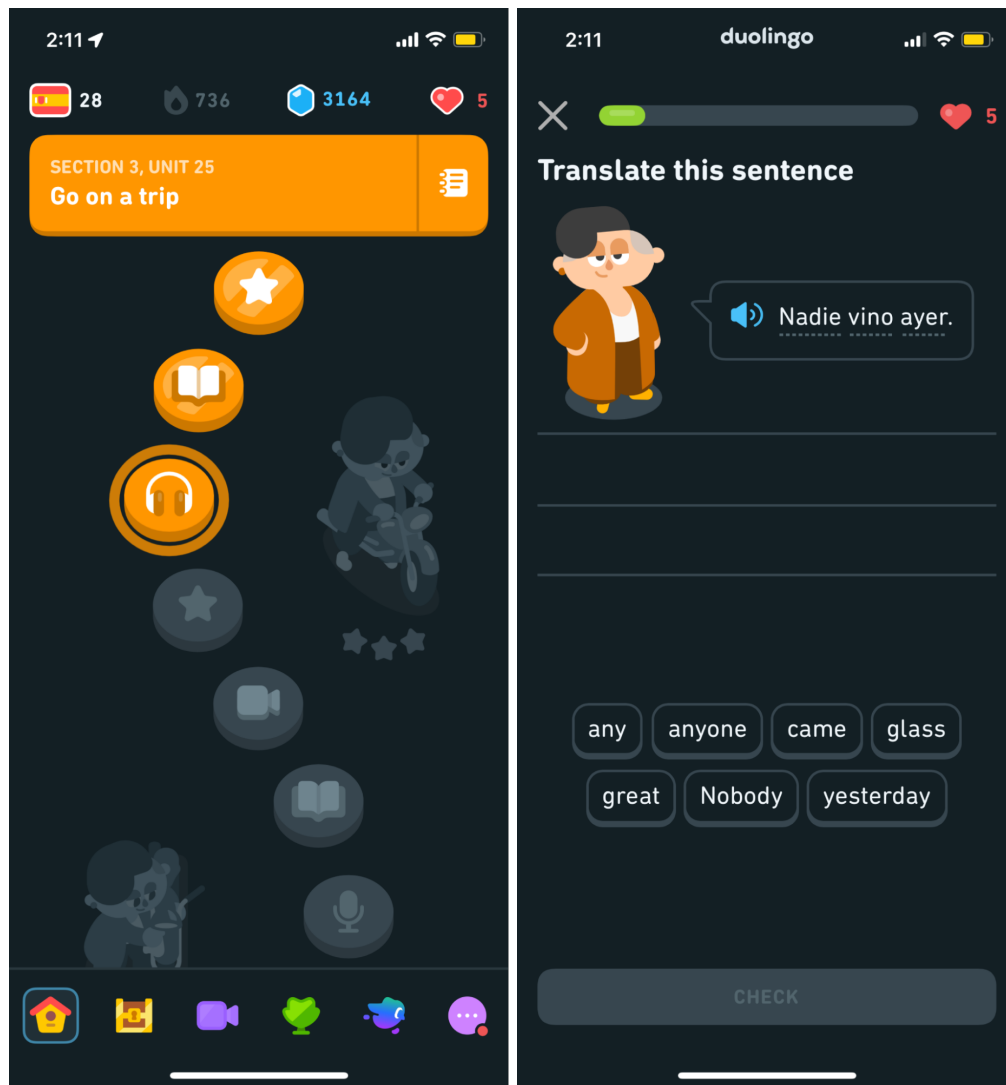


Figure 1: Two screenshots from an example Duolingo lesson (9/7/25). We took heavy inspiration from the layout that Duolingo uses to teach their users. We adapted their framework to fit better with both financial literacy, and a much smaller lesson set (three lessons).

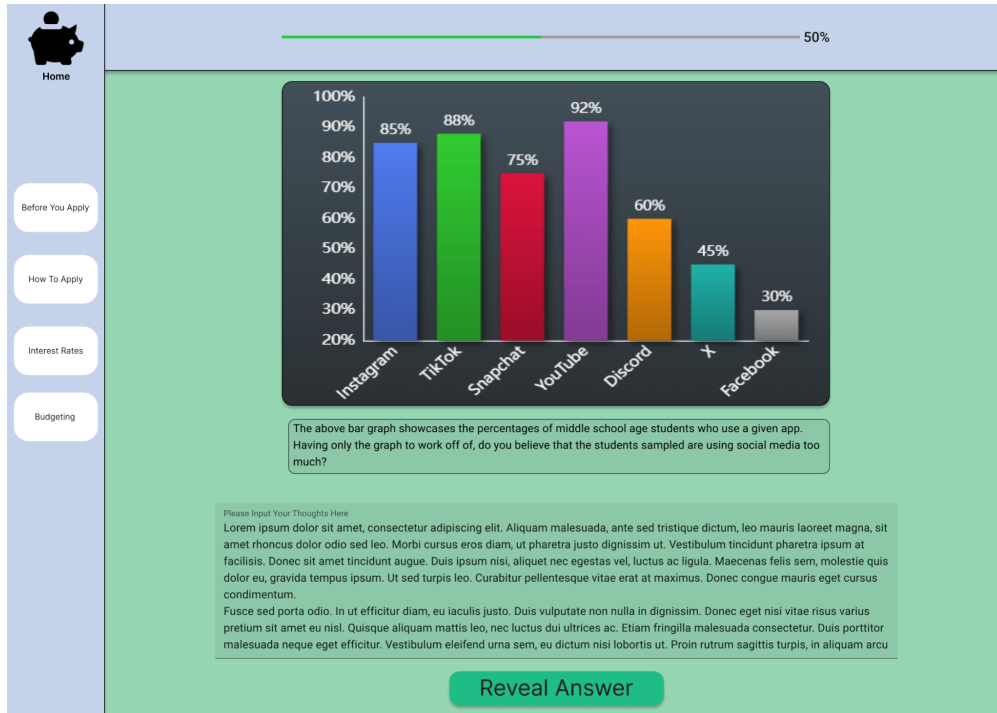


Figure 2: Our first mockup of what a page with a data visualization would look like. This initial mockup features many key features that we would continue to keep in our final app (after being redesigned). Examples of these are the navigational buttons, progress bar, piggy bank home icon, and the “Reveal Answer” button (9/2/25).

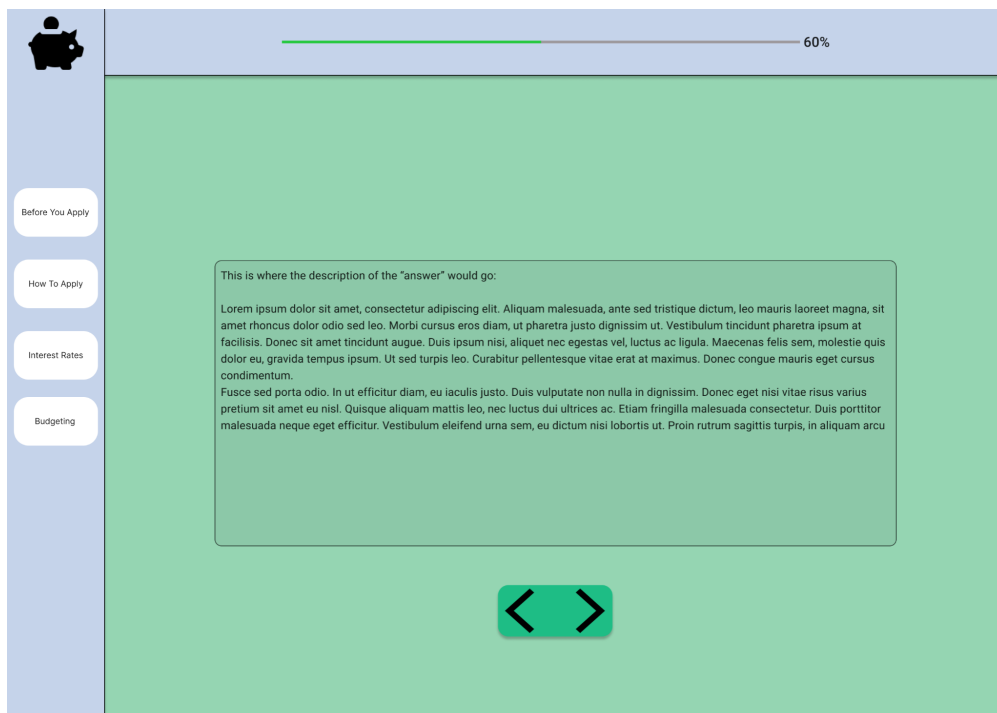


Figure 3: A continuation of the initial design talked about previously. This was our first idea for functionality of the “Reveal Answer” button (9/2/25).

Financial Literacy App

60%

Welcome back, Alex!

Continue where you left off or choose a new lesson.

Before You Apply



100%

How to Apply



60%

Interest Rates



20%

Budgeting



Not Started

Credit & Loans



Locked until
previous
lessons ∞

Saving & Investing



Locked until
previous
lessons ∞

Every step gets you closer to financial freedom!

Figure 4: The initial inspiration for the homepage of our site. This figure was produced by ChatGPT upon being given context of the idea and structure behind our site. While we did not use this exact structure, we took inspiration from its components and design choices (9/2/25).

3.1.3 Overall Structure

When first creating the content for our site, we decided we wanted to follow a format similar to Duolingo, where the user would first be greeted with a homepage displaying all the lessons they have completed, and the next lessons they should complete. Initially we wanted to have five total lessons, but decided to scale it down to three as our project took a turn into the process of AI implementation. Each of the three lessons has a similar, repeatable layout that has different content for each lesson. The general structure for each lesson is as follows:

Introductory Page: The user is greeted with some basic information about the given topic in a small paragraph. The user is also asked to answer a question that asks how proficient they believe they are on the given topic. This question is in the format of a likert scale. In some cases, the answer to this question will prompt a second question asking the user to rank the importance of subtopics within a given topic. **Subtopic 1 Introduction and Information:** This page may also contain an open ended question asking the user to think about a certain scenario. This is a sort of “pre-test” to get the user to start thinking about a topic before knowing all the information. The user is shown another short paragraph that introduces one of the subtopics within the selected topic. The user is also shown a visualization (sometimes interactive) that further teaches the user about the aforementioned topic. This visualization is meant to inform the user on what the subtopic is, why it is important, and (in some cases) how to perform the actions within the subtopic correctly. **Subtopic 2 Introduction and Information (If there is a second subtopic):** Same structure as the page above. **Comprehension Check:** The user is asked to answer four questions based on the material they were just presented. This test will help our team determine if the users are understanding all presented information. The basic question format is as follows (please note that questions may be presented in a different order): **Question 1:** The user is shown 2-4 brief scenarios of imaginary people and asked to determine which person is better adhering to the rules and principles discussed in the lesson. This question pertains to information discussed as subtopic 1. This is a multiple choice question. **Question 2:** The user is shown a question from the financial literacy exam [40][41] and asked to answer. This question pertains to general information about the overall topic. This is a multiple choice question. **Question 3:** The user is asked a question that relates to either subtopic 1 again, or subtopic 2. This question can have any of the following formats: open-ended, drag-and-drop, multiple choice.

Through our research into similar applications that strive to teach users to visualize data about a given topic, we determined that this was the most effective way to structure our lessons. This choice was further enforced by positive feedback to the overall layout on our first users tests.

After creating all the content for our three lessons in this format, our team decided to implement an AI coach. As of the time of writing this paper, artificial intelligence can be found in many applications, and several studies have been published claiming that AI helps users learn and understand topics more effectively [36] [37]. Even applications like Duolingo were switching to an “AI-first” approach where artificial intelligence was present in every step of the development and learning process. Our team decided to test this idea by adding an AI coach to our existing site and seeing if it helped or harmed users’ learning abilities. This will be further discussed in the “AI Integration” and “User Testing” sections below.

3.1.4 Figures of Final Site Design

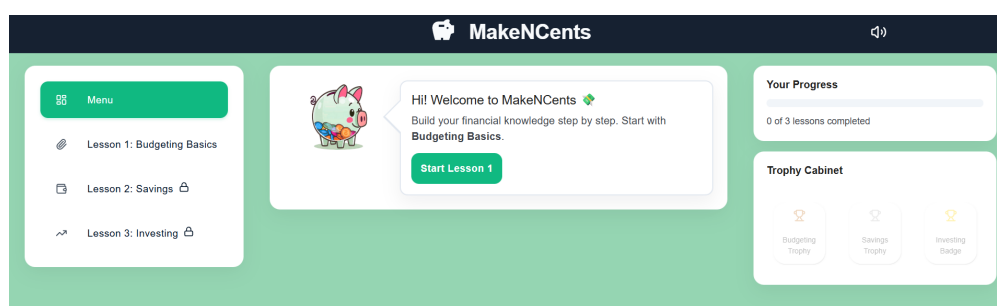


Figure 5: Final design for the homepage of our site. This is what the site looks like when a user first logs in, and has not previously completed any lessons.

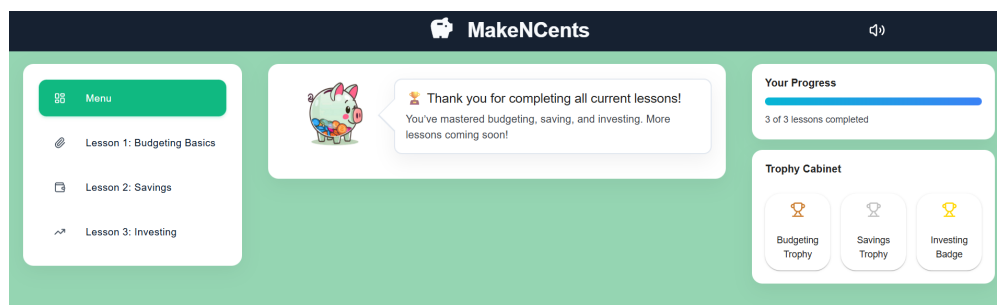


Figure 6: Final design for the homepage of our site. The user in the picture has completed all lessons.

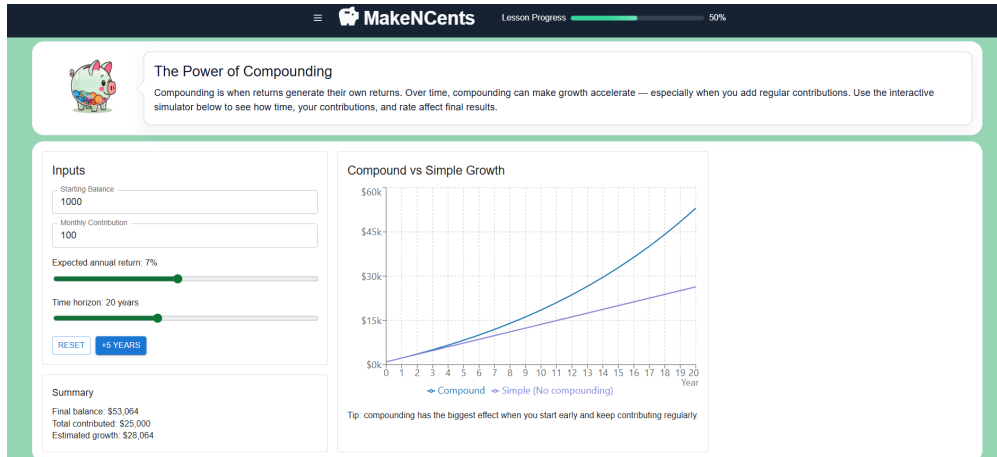


Figure 7: Example of a page with an interactive visualization. This page is about Compound Growth (Subtopic 1) in our Investing lesson.

Savings Comprehension Check

Question 1 of 3

Jim, Noel, Tracy and Molly are using different strategies to build their emergency savings fund. Who is NOT going to build an emergency savings fund quickly?

Jim	Noel	Tracy	Molly
No Look Savings	Impulse Purchasing	30 Day Savings	Pay Yourself First
Money directly deposited into their savings account without seeing it	Buys stuff without thought because they think they want/need it	Gives themself 30 days to think over an impulse purchase	Budgets aside a certain amount of their income to savings as a non-negotiable

Submit

Figure 8: Example of a comprehension check question. This screenshot comes from Question 1 of the comprehension check in our Savings lesson.

3.1.5 Kinetics and Engaging Features

Throughout our web application, we implemented a range of kinetic and interactive design elements to make the user experience feel more vibrant and rewarding. These animations and transitions were built primarily using “Framer Motion”, which is a React animation library that allows for smooth animations for visual components based on their status as well as user interaction. Our goal with these kinetics was to help users feel more engaged with their financial knowledge progress while also giving the application a sense of

fluid feedback.

A key feature of this is how every major element, whether that be navigation menus, lesson progress bars, and even question transitions, has been animated to subtly direct the user's focus. For example, the top navigation bar and menu dropdown use `motion.div` wrappers with fade-in and slide-down transitions (initial= opacity: 0, y: -20 and animate= opacity: 1, y: 0) so that the interface feels like it's gently coming into view as the user progresses through the application. When the user hovers over the hamburger menu, the `AnimatePresence` component renders the dropdown and animates it both on entry and exit. This prevents abrupt state changes and provides a polished, continuous flow.

Throughout the lessons, progress feedback is also affected by motion. The lesson progress bar at the top updates in real time as the user moves through different lesson parts, visually supporting their improvement. Buttons and interactive cards use `whileHover` and `whileTap` to produce scaling effects, such as slightly growing on hover (scale: 1.05) and shrinking on click (scale: 0.95). This mimics actual button feedback, enhancing the sense of responsiveness.

"Framer Motion" is also employed to create a conversational atmosphere between content sections. The Piggy Bank Mascot we use, for example, animates into view using spring scaling and rotation transitions (type: "spring", stiffness: 200). This animation gives the mascot a more active, playful style that makes the learning experience feel less static and more personable. Question prompts and visualizations also fade/slide into view using `AnimatePresence` with exit animations, making certain that each new part of the lesson flows smoothly from the last rather than appearing abruptly.

One last visually satisfying animation occurs at the end of the comprehension checks, when a checkmark is drawn progressively using "Framer Motion's" `pathLength` property. This kinetic "reward" ultimately gives the user a feeling of achievement and closure following the completion of each of the learning modules.

3.1.6 Sound Effects

When our group got the initial idea to make our web app, we took heavy inspirations from apps like Duolingo, that teach users complicated subjects in easy to understand and engaging ways. An aspect that we added to our project to mirror this engagement was the kinetics and other features described above. After our first complete prototype, we decided that adding sound effects to the comprehension checks might help the users feel more engaged, as Duolingo has many sound effects for getting questions correct or incorrect. We did this by finding two royalty free sound effects online (<https://pixabay.com/sound-effects/>). For the

incorrect sound, we picked a simple buzzer sound effect that would let the user know they got the question incorrect and they should try again. For the correct sound, we chose the sound of a coin being dropped into a metal piggy bank to let the user know when they got a question correct, while staying on theme with the rest of the website. After our second round of user testing, we got feedback that the sounds were much too loud for certain users. Because of this, we added a “Sound Settings” menu to the homepage where users can select how loud they want their sound effects to be, or turn them off completely. We also made it so all sound effects start at 50

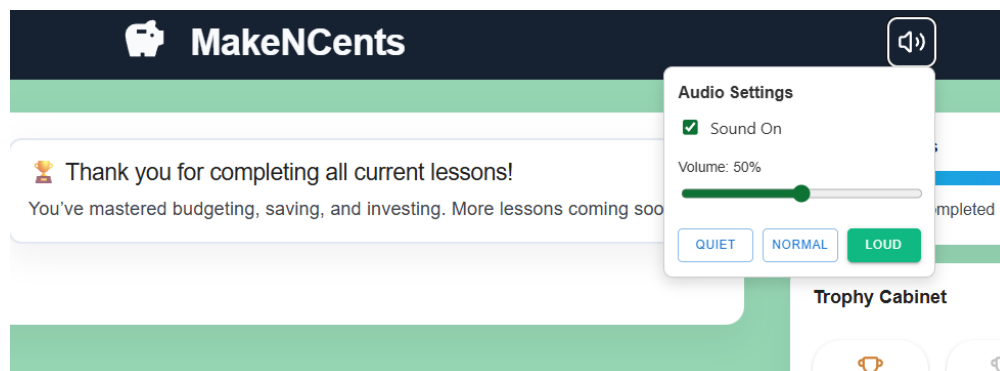


Figure 9: Our homepage with the “Audio Settings” menu opened. In this example, the user has their sound turned on and has selected the “Loud” preset.

3.1.7 Interactive Data Visualizations

Interactive visualizations were included in each learning module to help users learn the content and to add interactivity to the application. Additionally, the visualizations could be used to extract rough estimates of the users financial habits and data. For example, a user inputting their budget or savings data could give insights into the financial habits of the users. The visualizations were built using the D3 library.

The visualizations in the budget module include a stacked bar chart/donut chart for users to visualize their budget and spending across various categories. The savings module features a gauge visualization for users to input and visualize their progress towards building an emergency savings fund, and a line chart simulation to visualize different investment returns of different risk investments. The investment module includes a line chart with a diverging bar chart to compare the returns of a diversified stock portfolio and single stock portfolio and also includes a line chart to demonstrate the difference between compounding returns on investments over time, specifically in the long run.

Many of these visualizations include a wide range of interactive features where users can input their own data or simply change the scenario being shown. These visualizations are intended to help drive

further concreteness to the material and to make the learning experience more enjoyable, a key component to technology acceptance.

3.1.8 Locking and Unlocking Lessons

In the current iteration of our site, we have a feature that prohibits the user from starting Lesson 2 before completing Lesson 1. We believe that this feature helps guide the user in the most effective way, especially because future lessons will build on top of content from previous lessons. This makes it critical for the lessons to be completed in the designed order.

We accomplish this locking/unlocking of modules using a few different elements. Firstly, we initialize the state of all buttons past Lesson 1 (Lesson 2 and eventually onward) to be disabled by default. This makes it so that upon pressing the button, the user will not be directed to a new page before they have seen all previous content. We also use conditional statements to change the value of the words on each button when they are locked or unlocked. For example, when Lesson 2 is locked it reads, “Complete Lesson 1 First” and when unlocked it reads, “Lesson 2: Savings” (see comparison below). This helps give the user a clear understanding of why the button is disabled to start, and what steps they need to do to unlock the given button.

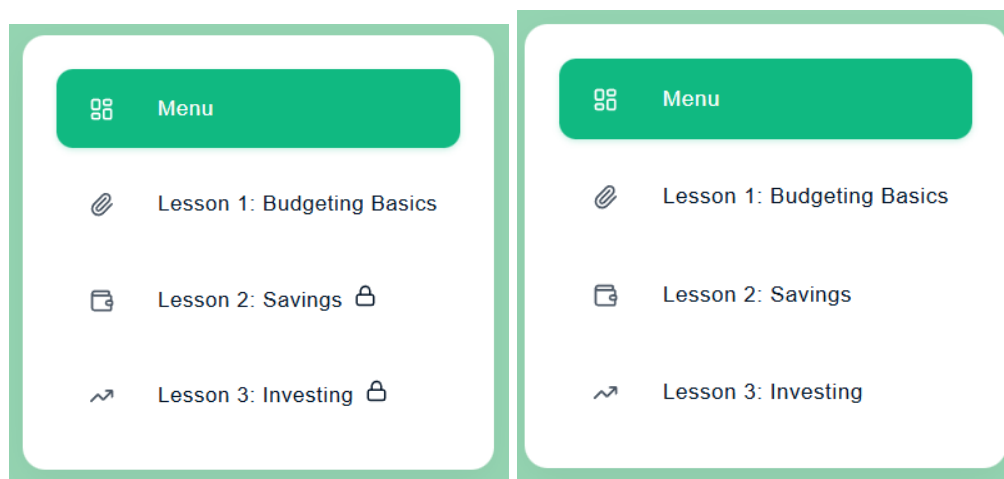


Figure 10: Comparison between all lessons locked (left) and all lessons unlocked (right)

Another element we use to accomplish locking/unlocking functionality is the use of a flag, given to the user upon completing each lesson. This flag is stored using local storage (cookies/cached browser data) so that it is carried between all pages of the site without the user needing to login or create an account.

These flags also persist across sessions, so each time the user closes and reopens the site, they will

still have access to their flags (as long as they do not clear their cached data). This works well for our current iteration; however, we may later adapt this approach to utilize GitHub authentication so that users can log in and receive a more personalized experience.

Luckily, the current codebase, which uses local storage, could be easily modified to handle authentication capabilities. Our homepage checks which flags a user is holding by looking for the `"lessonX_completed"` keyword in the user's local storage, and then sets a variable to `true` if it is detected. This allows the correct buttons to be enabled upon page reload.

This functionality is implemented using React's `useEffect()` hook. Please see the figures below for the specific code.

While we use cached data flags to make sure the user completes all lessons in the correct progression order, we also use these flags to give the user "Achievement Badges". Upon completing a lesson, the next time the user comes back to the homepage they will be greeted with a "Digital Medal" signifying that they have completed a lesson. Along with this they will be given a new prompt notifying them that the next lesson has been unlocked. We found this to be an effective method for making sure users do not unnecessarily complete the same lesson twice, a problem we encountered during our user feedback loop (talked about in the User Testing Section of this paper).



Figure 11: Initial mockup for what completing lessons would give the user. Please note the difference between a user that has completed no lessons (top) and all lessons (bottom). At first, we believed pixelated medals would be ideal, but we later changed this to fit the theme of our site better (12/11/25).

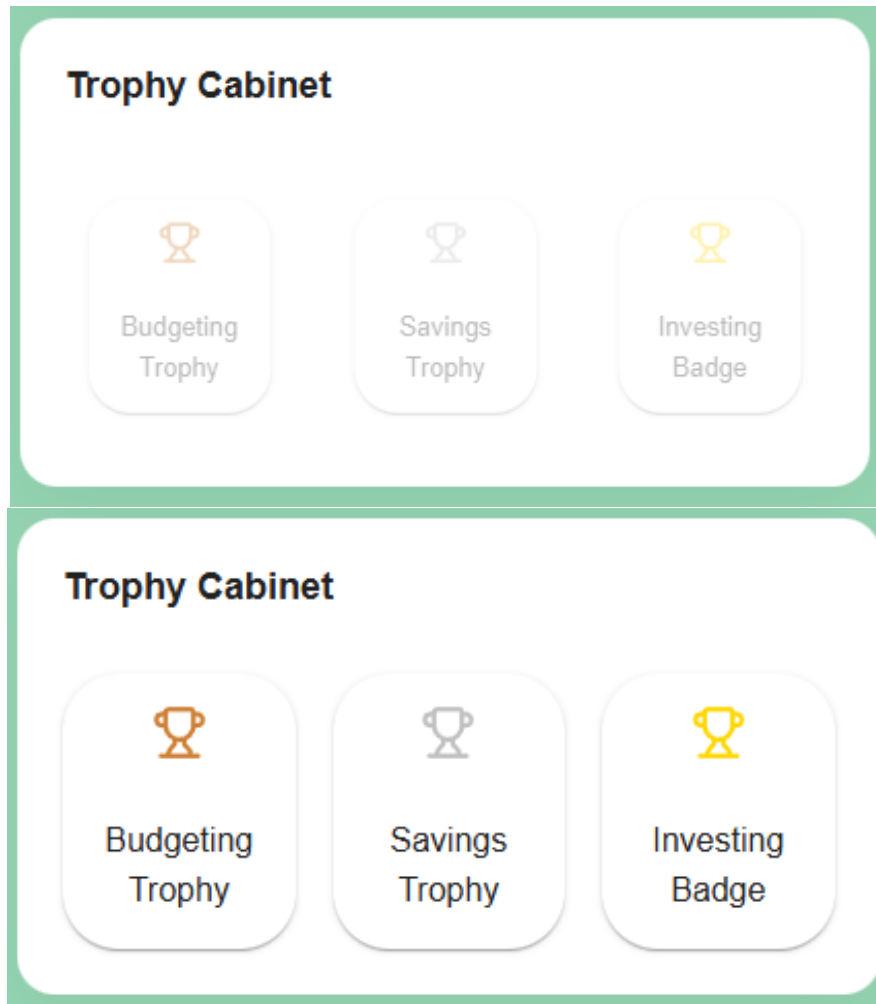


Figure 12: Final design for our “Digital Trophy Cabinet” that shows the user exactly which lessons they have completed. Please note the difference between a user that has completed no lessons (top) and all lessons (bottom).

3.1.9 AI Integration

In the current tech climate, artificial intelligence is in use by almost all major companies. Along with its increasing popularity, some studies have shown that the use of AI can help make learning easier for a broader population [36] [37]. After creating our initial three lessons worth of content, we wanted to experiment with ways that we could implement AI into our site to help facilitate a more personalized learning experience. The versatility of AI made it easy for us to brainstorm many different approaches to adding an AI agent to our site. All of the ideas we came up with can be found below.

3.1.9.1 Option 1: Tutorial Based Pacing

Overview: In this implementation of AI, we would leave the application’s structure mostly the same to how it is now, but wherever possible we would implement pages featuring an AI version of our site mascot that would introduce features or clarify possible misunderstandings. To help with this transition between AI pages and non-AI pages, we would have each interactive visualization on its own page.

Pros: This implementation would be most useful when it comes to our many visualizations as it would force the user to slow down and not unintentionally miss any features of interactivity. For example, our page featuring the “Build Your Own Budget” interactive chart would now be an AI agent guided experience with our mascot guiding the user into where they should try clicking and giving example values. Staying with the budget chart example, when the user first gets to the page, our mascot would pop up and the screen would be all greyed-out, except for the feature that the mascot is introducing.

Cons: With this implementation, interaction between the user and the AI chatbot is unavoidable, which may ruin the experience for users that do not want to have to be slowed down by the AI and just want to experience the site at their own pace.

3.1.9.2 Option 2: Unintrusive AI Helper

Overview: In this implementation of AI, the entire structure of the site would stay the same. AI would be implemented through a small icon of our site mascot in the corner. When this icon was clicked, a small chat window would open. This chat window would have a few recommended prompts, along with an open textbox where the user could come up with their own questions. Upon asking a question, a bigger chat window would open on the right side of the screen where the user could see the chatbot’s answer, while also being able to continue the conversation.

Pros: This implementation would be helpful to users who are confused about certain features of a visualization or the site itself, while also being completely optional and unobtrusive. This means that the pacing of the site would be entirely up to the user.

Cons: With this implementation, there is a high likelihood that some users would intentionally or unintentionally skip over the chatbot entirely, making it irrelevant to the site.

3.1.9.3 Option 3: AI Helper Sidebar

Overview: In this implementation, the structure of the site would remain very much the same, however there would be an added “AI Coach” button on the top left of the screen that the user could click to open a conversation window (similar to Github Copilot)

Pros: This option of the AI is a middle ground between ‘complete overhaul’ and ‘unobtrusive/optional’. The user can choose when and how much to talk to their AI companion making all users happy as they will be able to use or avoid it as much as they want.

Cons: Just as with previous options, this structure does not fully enable the AI to help the user if they are stuck unless the user initiates conversation first. This could cause parts of the site to be unintentionally skipped or looked over.

3.1.9.4 Option 4: ChatGPT Conversation Style

Overview: In this implementation, the structure of our site would be completely changed. There would no longer be the same three lesson progression structure. Instead, the user would be greeted with a full AI experience in which they would be prompted, answer questions, and the AI chatbot of our site mascot would bring up relevant information and visualizations for the user to look at and interact with.

Pros: This implementation would use the structure of successful chatbots like ChatGPT and Google Gemini to make learning easier and cleaner. The user would be able to move entirely at their own pace, while no longer having to complete lessons in order. This implementation would produce the most personalized experience, as every user would have a different path to learning the information.

Cons: Just as with the first implementation, this option may turn away people who are not comfortable conversing strictly with an AI chatbot. There is also a possibility that users may be less willing to give up estimates about personal information (such as monthly income and money spent on housing per month) as they will not be sure what the AI is doing with said information. Lastly, the AI will be in control of which lesson material to show in what order, which could cause a jumbled user experience if it were to jump around without a clear direction.

After discussing it between our group members and our advisor, we decided that Option 3: AI Helper Sidebar would be the best choice when it comes to integrating artificial intelligence into our site. We started first by creating a very basic sidebar conversation window that would appear on the left side of the screen at the user’s request. The AI agent within this sidebar has the ability to see that entire screen that

the user is on, along with all values that may be within the input boxes of our interactive visualizations. This allows the AI agent to answer specific questions the user may have about why visualizations are laid out the way they are. Once the foundations of this agent were in place, we chose to add character to the agent. We did this by giving the agent several prompts that would create a character centered around our website. The AI agent would represent our site’s mascot, a living piggy bank named Penny. This lets the users bond with the agent and creates a more personable experience.

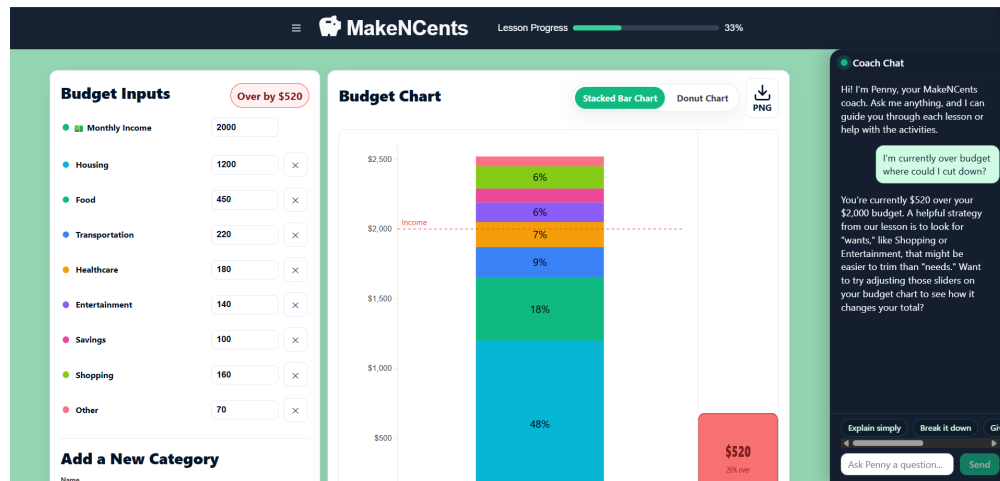


Figure 13: Example of a simple conversation with our AI coach. In this figure, the user asks Penny how they should interact with the visualization on screen.

3.2 User Testing

3.2.1 Revisit Study

To use ReVist in order to conduct a study, first the repo <https://github.com/revisit-studies/study> needs to be forked and cloned into. After forking and cloning the repo, a sample “.env” file is provided to make it easy to deploy using firebase. After setting up a new project in firebase for hosting, a few things need to be done in order for everything to work when changing “.env”.

1. VITE_BASE_PATH: Changed “/study/” to “/”
2. VITE_FIREBASE_CONFIG: Replace the provided config with your own from Firebase Console
3. VITE_STORAGE_ENGINE: Set to “firebase”
4. VITE_RECAPTCHA_V3_TOKEN: Set to your reCAPTCHA v3 site key (not secret key)
5. VITE_REPO_URL: Set to URL of your forked repo

After changing the “.env” file, all that needs to be done is logging in to firebase and deploying using the CLI. Lastly, if there are any CORS issues after deploying, they can be fixed via the Google Cloud SDK. After installing the SDK and setting it up using your google account, a “cors.json” file needs to be created as follows:

1. "origin": [Deployment URL(s)]
2. "method": ["GET", "HEAD", "OPTIONS"]
3. "responseHeader": ["Content-Type", "Access-Control-Allow-Origin"]
4. "maxAgeSeconds": 3600 (set to one hour)

To change the CORS policy using the Google Cloud SDK, run the following command:

```
gsutil cors set cors.json gs://BUCKET_NAME
```

To integrate our site in ReVisit as a study once it is fully set up, the first thing that needs to be done after deployment is creating a new directory within the root public folder. This new directory for our site is named “MakeNCents” and within this new “MakeNCents” directory, a “config.json” file needs to be included to detail the study’s structure. Within the “config.json” file the following items are included to initialize the study and include both screen and audio recordings:

1. "\$schema": (Use schema provided in examples)
`https://raw.githubusercontent.com/revisit-studies/study/v2.2.0/src/parser/StudyConfigSchema.json`
2. "studyMetadata": Includes the study’s title, version, authors[], date (yyyy-mm-dd), description, and organizations[]
3. "uiConfig": Includes the study’s logoPath, contactEmail, and four boolean variables
Required config variables: [withProgressBar, withSideBar] (currently set to false)
Screen/audio recording variables: [recordScreen, recordAudio] (currently set to true)
4. "importedLibraries": ["screen-recording"]
5. "components": Includes all necessary components. For the MakeNCents site, only one component is required to load a website.
6. "task": Includes the study’s type (website), path (external URL or local path to build assets), withSideBar (boolean), and response[] (currently set to an empty array)

7. "sequence": Determines the order in which components are shown to the participant. Structured using "order": fixed, with components listed underneath as:
`"$screen-recording.co.screenRecordingPermission", "task"`

3.2.2 Firebase Storage

When completing user testing we would record the screens and audio of each participant to understand how they interacted with the site. These recordings would be automatically done through the Revisit site, once the user consented to being recorded. When setting up firebase for hosting and authenticating in Revisit, storage was also set up through firebase for screen recordings. Every participant that consented to being recorded would then have their video automatically uploaded to firebase storage as soon as the study ended. We found this to be the easiest way to store the videos as Revisit already had a provided setup for using firebase storage that just needed to be set up for our personal use.

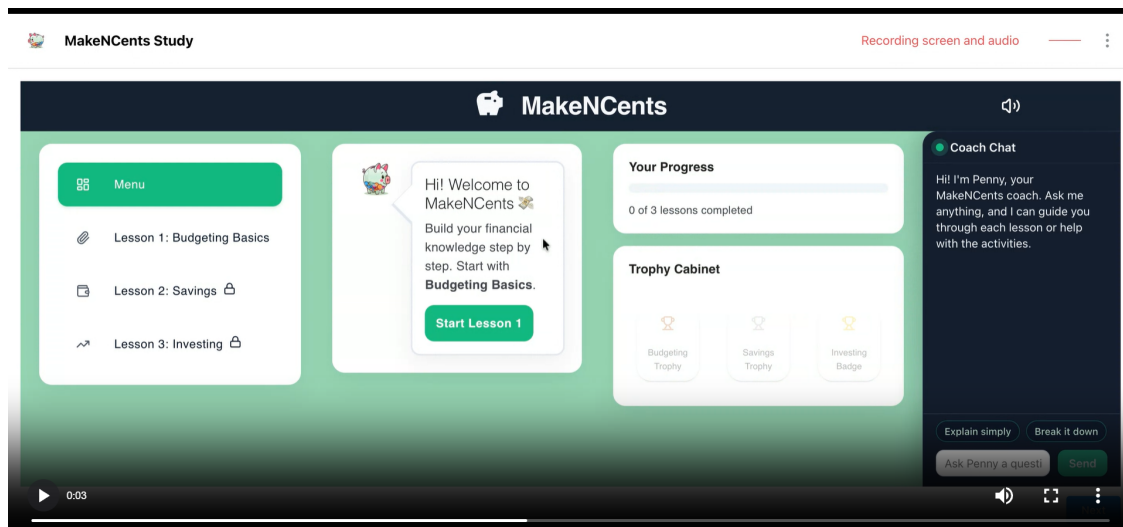


Figure 14: Example recording from our firebase storage

3.2.3 Study Specifics and User Accessibility

To run the studies, we decided on using the platform Prolific to find participants online. We decided to run the study on 18–25 year olds that lived in the United States with an equal number of participants from both male and female. We chose this age range as we believed our financial modules would help in educating young adults the best. Initially, for our first prototype, we instructed the Prolific participants with the following text:

In this study, you will complete a short interactive budgeting lesson on an external website. The lesson has four parts:

- 1. A brief question about your current budgeting habits.*
- 2. A short explanation of a budgeting rule with a simple visual.*
- 3. An interactive budgeting chart where you enter your monthly income and explore how it is spread across multiple categories.*
- 4. A short comprehension check with a few quick questions.*

As you work through the lesson, you will be asked to think aloud by talking through what you are doing, noticing, or finding confusing while using the site and answering the questions.

The study should take anywhere from 10–15 minutes to complete.

After reading the instructions, the participant would then click a link bringing them to our hosted Revisit site, allowing them to run the study and test our site. Over time, these instructions were refined, as sometimes, participants did not seem to speak aloud as they went through the site or had no clue at first what exactly the site was for. The new refined instructions ended up being the following text:

In this study, you will complete a short interactive budgeting, savings, and investing lesson.

We ask you to please “THINK ALOUD” by talking through what you are doing, noticing, or finding confusing while using the site. This does not have to be constant, but we appreciate any input!

Once on the site, click the “go to study” button and allow screen and audio recording.

The study should take anywhere from 10–20 minutes to complete. Taking more time is also more than welcome!

Lastly, when conducting A/B testing later on, an additional tag was added for those who were randomly selected to use the AI coach. The tag added is the following text:

There is a chance an “AI Coach” named Penny will be available to support you in any way necessary, which we please ask you to use if you find anything confusing or want further explanations on any part.

Overall, we found that including the “THINK ALOUD” portion in all capital letters helped participants easily see what it was we were looking for from the study. Additionally, removing the numbered list

and keeping it to short sentences seemed to be effective in getting instructions across to more participants in full.

3.2.4 User Testing Loop

Once we had finalized the framework we wanted to use for our user testing, we began to discuss what format we wanted to use to get the most effective information. We first decided to run a Revisit Study that contained only the first two lessons (Budgeting and Savings) to make sure that everything was running smoothly. We ran our first study with one participant. After viewing the study video, we found that this user was unable to complete the lesson due to some viewport issues (mainly the Study Browser bar blocking a lot of page content). We made several fixes to ensure that the user would be able to actually attempt the content on the page, then we ran a second study, this time with two participants. This study went much better, resulting in both participants being able to complete both lessons. While watching the screen recordings from the study, we took notes on many small changes that we intended to make. We continued to do this cycle (run test, watch video, take notes, make changes, repeat) while continuing to finish the content for our third lesson (investing). Once the third lesson was complete, it was also added to the study so users could not experience the entire site's content. Upon doing this we continued to run our cycle with another 3 participants, this time gaining some very valuable information on features we should change and features we should add as delighters. We changed our cycle slightly upon having all the content finished, now we would take notes on the screen recordings while watching, then we would go back through and color code each note according to a scale that we determined as a team. We would color code as follows:

- **Red** = Things that need to be changed
- **Yellow** = Things to keep an eye on and may need to be changed in the future
- **Green** = Things that worked well or that users liked

After changing to this scale, we found it much easier to prioritize the changes that were greatly affecting the user, while also adding delighter features or tweaking small things to try and improve our user experience. This included things like adding sound effects (and a sound settings menu for those sound effects), giving our site mascot a name and making them more personable, making certain buttons stand out more so that the user would be more inclined to interacting, changing the viewing size of the screen and its components, and many more. A breakdown of our user testing template can be found in Appendix A1.

During this time in our development, we decided to split the team up into two smaller teams. One

team would continue our user testing loop on our site with its three completed lessons. The other team was starting to develop and test what our site would look like with our AI coach integration that was discussed above. We chose to split up this way so that we could more effectively test the site in an A/B testing format. We decided to perform two concurrent studies so that we could try to determine if people learned more or less when helped by an AI coach. In our test, Option A was the original site with all three completed lessons and no AI coach, while Option B was the site with the same content while also being integrated with our AI coach sidebar and tooltips. To attempt to determine which option was performing better on understanding the visualizations, we added an additional question to the comprehension check at the end of each lesson. We believed that if the users could accurately recall these visualizations and what they were meant to be used for, then they would have fully understood that visualization.

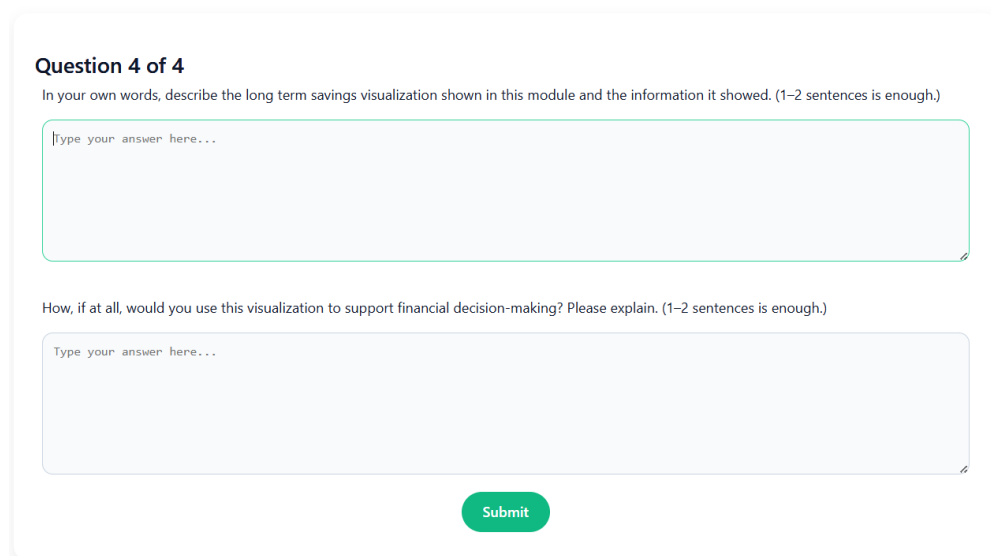
The image shows a digital form for a comprehension check. At the top, it says "Question 4 of 4". Below this is a prompt: "In your own words, describe the long term savings visualization shown in this module and the information it showed. (1-2 sentences is enough.)". There is a large text input area with a light blue background and a green border, containing the placeholder text "Type your answer here...". Below this is another prompt: "How, if at all, would you use this visualization to support financial decision-making? Please explain. (1-2 sentences is enough.)". There is a second, identical text input area with the placeholder "Type your answer here...". At the bottom center of the form is a green "Submit" button.

Figure 15: Example of what the fourth question on each comprehension check looks like

With the users answers from these questions in both Option A and Option B, we could come together as a team and determine which users fully understood the visualizations they were faced with, and whether an AI coach helped or harmed learning. Our exact quantifiers for determining which option was better can be found in the Results section. Below is a figure of our user testing process simplified.

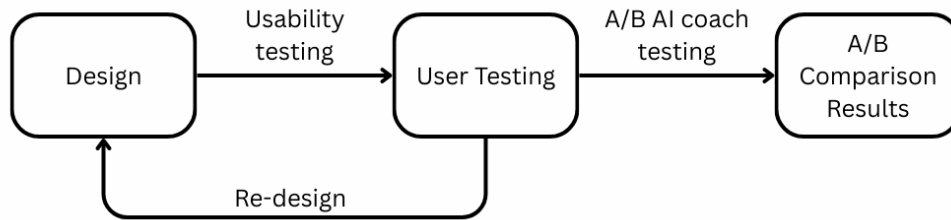


Figure 16: Our user testing process (simplified)

3.3 AI Coach

3.3.1 Why Implement an AI Coach?

We initially came up with the idea of implementing an AI coach for our web application to enhance user engagement and to make the learning experience more interactive and personalized. By integrating AI into their experience, users have the opportunity to more actively explore the provided modules and data visualizations through added guidance and feedback that can assist with their individual needs and work at their particular learning speed. The AI coach also provides a more conversational experience for the user, allowing them to ask questions and receive explanations in a dialogue-based format rather than passively consuming the information provided in our modules. It can serve as a supportive assistant whenever the user is confused or needs help. This feature also allowed us to test the effectiveness of generative AI in a learning environment, providing us with insight into how such technology can support financial literacy and improve overall understanding.

3.3.2 AI Coach Overview

The AI coach is active on every main page of our site. The coach is initially set up with information about the site in general. Additionally, the coach is state-aware by obtaining a curated snapshot of the specific parts of the page state that matter, for the current page the user is on. This snapshot includes the user's overall module completion progress, current lesson progress, and page-specific interaction data the user may have changed, including any interactive visualizations or answers to comprehension checks. Along with this snapshot of the site, the coach is provided with a custom system prompt focusing on giving users concise and supportive guidance. This guidance was set to be hint-first and approach helping the user by going one step

at a time instead of trying to push ahead. We also included some conversational presets such as “Explain simply,” “Break it down,” “Give an example,” and “How do I use this?” to encourage initial engagement and get users to easily begin conversing with the AI coach. The chosen model for this coach was Gemini 3 Flash Preview, primarily for its fast response time and overall strong performance.

3.4 Analysis Methods

3.4.1 A/B Testing Overview

In order to analyze the perceived helpfulness of incorporating an AI Coach in our web application, we performed an A/B testing study to evaluate user experience with and without the AI coach (Option A: AI coach; Option B: No AI Coach). For this study, we developed a scoring schema to rate each participant based on their overall engagement throughout each module and how they performed during the comprehension check portions of the modules. The criteria in our scoring schema were graded out of 1, 3, or 5 points depending on the category. A user that achieved full marks in each criteria would get a score of 30.

Table2									
Version	Coach Engagement	Date	Participants	Engagement Budgeting	Engagement Savings	Engagement Investing	Budgeting Q1-3		
Coach Pilot	None	2/16/26	Participant 1	4	4	3	3		
Coach Pilot	None	2/16/26	Participant 2	5	2	3	3		
Coach Pilot	None	2/16/26	Participant 3	0	0	0	0		
Budgeting Q4.1	Budgeting Q4.2	Savings Q1-3	Savings Q4.1	Savings Q4.2	Investing Q1-3	Investing Q4.1	Investing Q4.2	Total	
1	0	2	0	0	3	1	0	21	
1	1	3	1	1	3	1	1	25	
0	0	0	0	0	0	0	0	0	

Figure 17: Examples of graded user results. Results from our first three users from the AI coach version of our site.

3.4.2 A/B Scoring Schema Breakdown

When brainstorming ideas for how to grade users, our team decided that we wanted to grade users on how engaged they were with the site. We came to this decision because we believed that if a user was more engaged with the site and its features, they would learn and understand more of the presented information. Our team then decided to come up with a way to quantify how engaged a user was using the system below:

3.4.2.1 Engagement Scoring System

Maximum: 5 Points

- **0–3 Points:**

- User did not rush through the lesson
- User took time to read all information
- User spent time reading and interacting with all visualizations

- **0–2 Points:**

- User connected the information learned during the lesson to their own personal finances

3.4.2.2 Qualitative Data (Categorical)

Version: Indicates which version of the site the user interacted with. Possible options are “Coach” (Option A) and “No Coach” (Option B). The addition of “Pilot” indicates first-round test users.

Coach Engagement: Measures how much the user engaged with the AI coach. This does not apply to “No Coach” (Option B) users. Possible values: “None”, “Some”, “A Lot”.

Date: The date the participant completed the study. This helps determine differences across testing days.

Participant: A numbered index of users from the same day, beginning at 1.

3.4.2.3 Quantitative Data (Numerical)

Budgeting Engagement: Engagement score for the Budgeting Lesson (Lesson 1). See Engagement Scoring System above.

Savings Engagement: Engagement score for the Savings Lesson (Lesson 2).

Investing Engagement: Engagement score for the Investing Lesson (Lesson 3).

Budgeting Q1–3: First three comprehension questions for Budgeting (Lesson 1). Users earn 1 point per correct first attempt (max 3 points).

Budgeting Q4.1: First part of Question 4 (Budgeting). 1 point if the response accurately recalls the referenced visualization.

Budgeting Q4.2: Second part of Question 4 (Budgeting). 1 point if the response explains how the visualization applies to the user’s personal finances (e.g., using “I”, “Me”, or “Mine”).

Savings Q1–3: First three comprehension questions for Savings (Lesson 2). 1 point per correct first attempt (max 3 points).

Savings Q4.1: First part of Question 4 (Savings). 1 point for accurate visualization recall.

Savings Q4.2: Second part of Question 4 (Savings). 1 point for personal application explanation.

Investing Q1–3: First three comprehension questions for Investing (Lesson 3). 1 point per correct first attempt (max 3 points).

Investing Q4.1: First part of Question 4 (Investing). 1 point for accurate visualization recall.

Investing Q4.2: Second part of Question 4 (Investing). 1 point for personal application explanation.

Total Score: Total cumulative score across all categories. A perfect score is 30 points.

3.4.3 A/B Testing Checklist

To assist us with deciding scores for each participant for each quantitative category in our scoring schema, we developed a comprehensive checklist that details the main factors that we wanted to see the participants interact with or achieve. Participants would receive points for how well they aligned with our checklist. Each module had similar checklist categories for the comprehension check portions that prioritized correctness, as well as thoughtfulness and personalization in the open-response prompts. Each module also had some unique checklist categories based on their contents focusing on overall module understoodness (which was partially influenced by whether they spoke aloud), as well as detailed data visualization interaction and personalization.

Module	Observation / Task
General	Date
General	Think-Aloud: Did they talk through their process? (Note frustration)
General	AI Coach: Did they hover, scroll presets, or ask questions?
General	Tech/UI Issues: Note bugs, responsive issues, or illegible text
General	Direct Quotes: Notable quotes about UI or experience
Budgeting	Intro: Did they read/understand the text and visual aids?
Budgeting	Chart: Did they modify "Monthly Income" to reflect reality/scenario?
Budgeting	Chart: Did they edit, delete, or add budget categories?
Budgeting	Chart: Did they toggle viz options (Bar vs. Donut)? Preference?
Budgeting	Chart: Did they ensure the budget balanced (no unallocated money)?
Budgeting	Comp Check: Number of attempts to get all questions correct
Saving	Intro: Did they read/understand the information?
Saving	Dial: Did they interact with the Savings Dial or skip it?
Saving	Dial: Did they input their own monthly income and current savings?
Saving	Dial: Did they expand and read the "Savings Tips"?
Saving	Risk: Did they click "Learn More" on the risk options?
Saving	Risk: Did they explore 1yr, 5yr, 10yr horizons on the graph?
Saving	Risk: Did they read the "Illustrative Decrease Explanations"?
Saving	Comp Check: Number of attempts to get all questions correct
Investing	Intro: Did they read/understand the information?
Investing	Stock Sim: Did they hover over graphs to read tooltips?
Investing	Stock Sim: Did they experiment with different stock options?
Investing	Stock Sim: Did they change the Investment Amount?
Investing	Compounding: Did they interact with the data sliders?
Investing	Compounding: Input Validation (e.g., clearing the initial '0')
Investing	Compounding: Edge Cases (e.g., inputting extremely high numbers)
Investing	Compounding: Did they click preset scenarios (e.g., "Save for education")?
Investing	Comp Check: Number of attempts to get all questions correct
Q4 Responses	Budgeting Prompt 1 (Exact Quote)
Q4 Responses	Budgeting Prompt 2 (Exact Quote)
Q4 Responses	Saving Prompt 1 (Exact Quote)
Q4 Responses	Saving Prompt 2 (Exact Quote)
Q4 Responses	Investing Prompt 1 (Exact Quote)
Q4 Responses	Investing Prompt 2 (Exact Quote)

Figure 18: Detailed contents of our checklist for each module in our web application as well as some other general information we were looking for from participants in our study

3.4.4 A/B Checklist Breakdown

3.4.4.1 General

Date: The date that a given user participated in our study through ReVisit.

Think-Aloud: Did they talk through their process? (Note frustration) This item is for if a user explained what their thought process was as they were going through the web application. Did they explain what they were doing? Did they explain why?

Tech/UI Issues: Note bugs, responsive issues, or illegible text This item is for detailing any visible technical issues seen by the reviewer of a given user screen recording video, or any technical issues mentioned by the participant that can be fixed in the future.

Direct Quotes: Notable quotes about UI or experience This item is for any helpful information or feedback provided by the participant.

3.4.4.2 Budgeting

Intro: Did they read/understand the text and visual aids? This item is for noting whether it appeared the participant was understanding the information provided to them in the introduction of the budgeting module.

Chart: Did they modify "Monthly Income" to reflect reality/scenario? This item is for noting whether the participant modified the "Monthly Income" portion of the budgeting chart with their own information or a made up scenario instead of leaving it as the default.

Chart: Did they edit, delete, or add budget categories? This item is for noting if the participant tried editing existing budgeting chart categories or adding new categories.

Chart: Did they toggle viz options (Bar vs. Donut)? Preference? This item is for noting if the participant tried changing between the bar chart and donut chart options for the budgeting chart.

Chart: Did they ensure the budget balanced (no unallocated money)? This item is for noting if the participant ensured they allocated 100% of their monthly income on the budgeting chart.

Comp Check: Number of attempts to get all questions correct This item is for noting how many attempts it took the participant to get the first 3 questions in the budgeting comprehension check correct.

3.4.4.3 Savings

Intro: Did they read/understand the information? This item is for noting whether it appeared the participant was understanding the information provided to them in the introduction of the savings module.

Dial: Did they interact with the Savings Dial or skip it? This item is for noting if the participant interacted with the Savings Dial in the savings module.

Dial: Did they input their own monthly income and current savings? This item is for noting if the participant entered unique information into the monthly income and current savings options for the Savings Dial instead of leaving those options as default.

Dial: Did they expand and read the "Savings Tips"? This item is for noting if the participant read the "Savings Tip" portion on the Savings Dial page of our savings module.

Risk: Did they click "Learn More" on the risk options? This item is for noting if the participant clicked on and read the "Learn More" options in the risk section of the savings module.

Risk: Did they explore 1yr, 5yr, 10yr horizons on the graph? This item is for noting if the participant interacted with and explored the 1 year, 5 year, and 10 year horizon options on the savings risk graph.

Risk: Did they read the "Illustrative Decrease Explanations"? This item is for noting if the participant read the "Illustrative Decrease Explanations" portion of the savings risk graph.

Comp Check: Number of attempts to get all questions correct This item is for noting how many attempts it took the participant to get the first 3 questions in the savings comprehension check correct.

3.4.4.4 Investing

Intro: Did they read/understand the information? This item is for noting whether it appeared the participant was understanding the information provided to them in the introduction of the investing module.

Stock Sim: Did they hover over graphs to read tooltips? This item is for noting whether or not the participant hovered over the tooltips in the investing stock visualization.

Stock Sim: Did they experiment with different stock options? This item is for noting whether or not the participant tried out the different stock options provided in the investing stock visualization.

Stock Sim: Did they change the Investment Amount? This item is for noting if the participant changed the investment amount in the investing stock visualization.

Compounding: Did they interact with the data sliders? This item is for noting if the participant tried adjusting the data sliders in the compound interest visualization.

Compounding: Input Validation (e.g., clearing the initial '0') This item is for noting if the participant tried inputting unique information into the compound interest visualization.

Compounding: Edge Cases (e.g., inputting extremely high numbers) This item is for noting if the participant tried entering edge case values into the compound interest visualization.

Compounding: Did they click preset scenarios (e.g., "Save for education")? This item is for noting whether the participant clicked on the preset scenarios presented at the bottom of the compounding section.

Comp Check: Number of attempts to get all questions correct This item is for noting how many attempts it took the participant to get the first 3 questions in the investing comprehension check correct.

3.4.4.5 Q4 Responses

Budgeting Prompt 1 (Exact Quote) The participant's exact response to Question 4 Part 1 of the budgeting comprehension check.

Budgeting Prompt 2 (Exact Quote) The participant's exact response to Question 4 Part 2 of the budgeting comprehension check.

Saving Prompt 1 (Exact Quote) The participant's exact response to Question 4 Part 1 of the savings comprehension check.

Saving Prompt 2 (Exact Quote) The participant's exact response to Question 4 Part 2 of the savings comprehension check.

Investing Prompt 1 (Exact Quote) The participant's exact response to Question 4 Part 1 of the investing comprehension check.

Investing Prompt 2 (Exact Quote) The participant’s exact response to Question 4 Part 2 of the investing comprehension check.

3.4.5 A/B Testing Process

After properly organizing our checklist and user scoring schema, we began the process of collecting participant data. Similarly to how we found participants for our initial user testing of our web application, we once again used ReVisit to get participants for our A/B testing. ReVisit allowed us to collect screen recordings of our participants interacting with our web application, providing audio input capabilities for the user to talk aloud. Our participants were all from the United States, were aged between 18 and 25, and they had a 50/50 gender split.

We performed two iterations of A/B tests, each having 6 participants with the AI coach and 6 without, for a total of 24 participants (plus one extra participant in the non-AI coach group who was added unintentionally). For each iteration, two of our team members would view each participant’s screen recorded run through of the web application, and score based on the checklist and schema. That way, each participant submission was reviewed twice and teammates who reviewed the same recordings discussed and agreed on what scoring made the most sense.

3.5 Data Processing

Finally, data was cleaned using R and stored in an excel file. Importantly, additional processing of the data included flagging users who did not complete all learning modules and filtering out users who did not complete the budgeting module (n=2). This decision was made because those users’ results would be unhelpful in our analysis due to their incompleteness. This left us with a final dataset of 12 coach users and 11 non coach users (we accidentally had an extra coach user upon initial testing prior to filtering).

4 Results

This section presents the results of our A/B testing comparing how much users learned and engaged with the site based on what version of the site they used, as well as some other key findings.

4.1 Learning Outcomes

Here, we analyze the learning outcomes of our non-coach and coach users. To analyze learning outcomes we used users' self assessment of their financial literacy both before and after going through our learning modules. We noticed no difference in learning outcomes for users who had the AI coach vs users who did not have the AI coach.

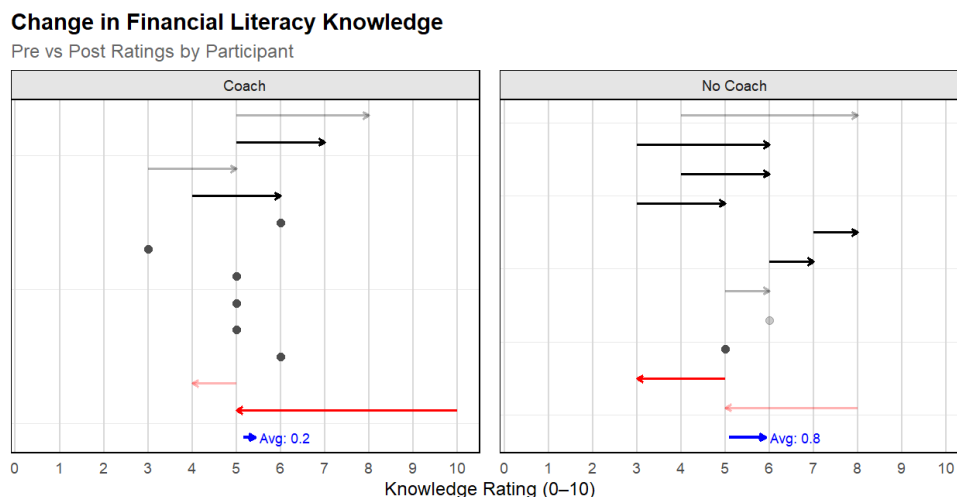


Figure 19: Change in financial literacy ratings before and after our modules comparing the coach and non-coach version.

Figure 19 shows how much each user's financial literacy improved or decreased after going through the learning modules. The beginning of each arrow indicates their initial self-assessment while the end indicates the post self-assessment where color indicates an increase or decrease. Dots indicate no change. On average, users with the AI coach said their financial literacy improved by 0.2 while people without the coach said it improved by 0.8. Notably, the unshaded arrows and dots indicate users who did not complete all learning modules (n=3 for both the coach version and non-coach version for a total of n=6).

Many of the users with the coach version reported no change (n=5), compared to the non-coach version only a couple reported no change (n=2). Overall, the results based on pre and post financial literacy

assessment indicate that the AI coach had no impact on improving learning outcomes.

4.2 Engagement

In this section, we analyze the engagement scores between our two groups. To analyze the engagement of both versions, we used the total scores of our engagement scoring system (“See A/B Scoring Schema Breakdown”). This system scored users on a scale of 0-30 considering their overall engagement, use of learning materials in the modules, and their scores on the comprehension check of modules. In general, we found only a small difference in engagement between the versions.

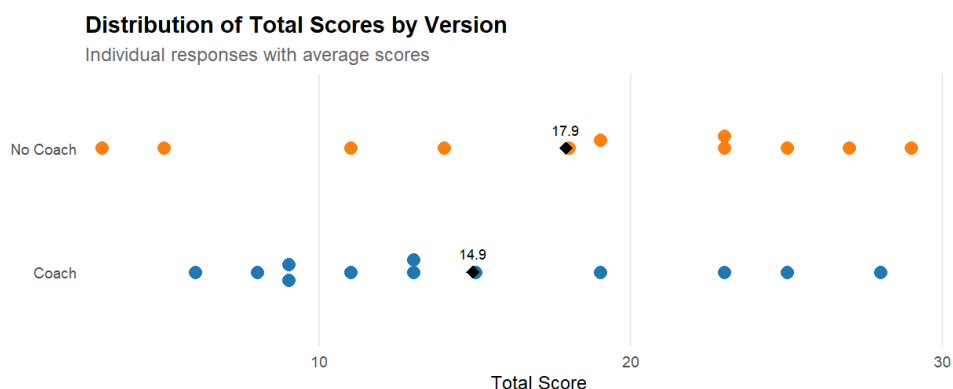


Figure 20: Engagement ratings for coach vs non-coach users

In figure 20 we compare the engagement scores of users’ with the AI coach (blue) and without it (orange). The non-coach users had an average engagement score of 17.9 and the coach users had an average score of 14.9. The non-coach users had a higher spread of score with the lowest score being a 3 and the highest being a 29. For the coach users the scores were distributed tighter with a lowest score being a 6 and highest being a 28.

We also saw more non-coach users scoring above 20 ($n=5$) compared to coach users ($n=3$). In all, we saw a small difference in engagement of the coach and non-coach users, with only slightly more engagement for non-coach users.

4.3 Data Visualization Impact

To assess the impact of the data visualizations on financial literacy learning, we examined the data visualization interactions of the users. Using the scoring schema to determine how often the participants interacted with the data visualizations, we ended up with 3 groups. Those with low interaction (0-4 inter-

actions with the data visualizations), medium interaction (4.5-9.5 interactions), and high interaction (10+ interactions). 9 participants ended up in the low interaction group, 7 in medium, and 7 in high.

When looking back at the self-assessment financial literacy rating entered by participants, we noticed that participants gave improvement to their financial literacy rating with more data visualization interaction. Those who actively engaged with the data visualizations and were marked as having “high interaction”, did see an average of +1 financial literacy rating between the pre and post survey in comparison to those who had medium to low interaction, which saw little to no change in financial literacy ratings. This indicates that the visualizations may help participants a little when it comes to their own self-confidence in financial literacy, but it does not clearly say that visualizations strongly improve financial literacy ratings.

4.4 Module Results

In this section, we analyze the results from each module based on engagement and performance on comprehension checks to see if certain modules/topics were particularly engaging or hard for users to learn. We found that users’ engagement tended to trail off throughout the learning experience.

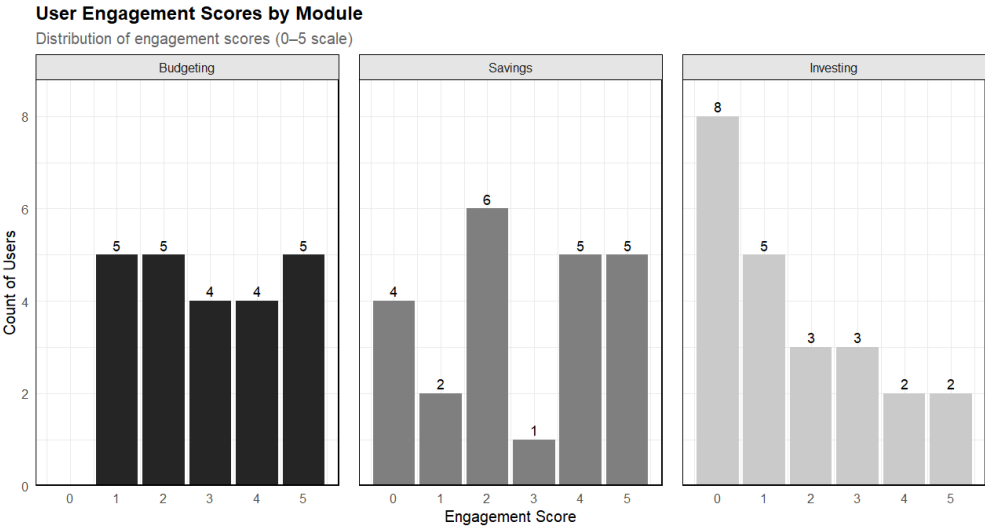


Figure 21: Engagement scores for each module

In figure 21 we compare users’ engagement scores for each particular module with no distinction for which version they had. These scores were given on a scale of 0-5 and scored based on how much they interacted with key parts of each learning module. From the figure, it can be observed the engagement in the Budgeting module (1st module) was somewhat spread out but mostly greater than 2 (n=13). As we move to the Savings module (2nd module) scores begin to be more spread out with users either being very engaged

or not engaged. Finally, as users got to the Investing module (3rd module), engagement severely declined with a majority of users earning a 0 or 1 (n=13) and only a few earning a 4 or 5 (n=4). This suggests that users' attention or enjoyment throughout the learning experience seemed to trail off as they moved through the modules with the last module (Investing) having the lowest engagement scores.

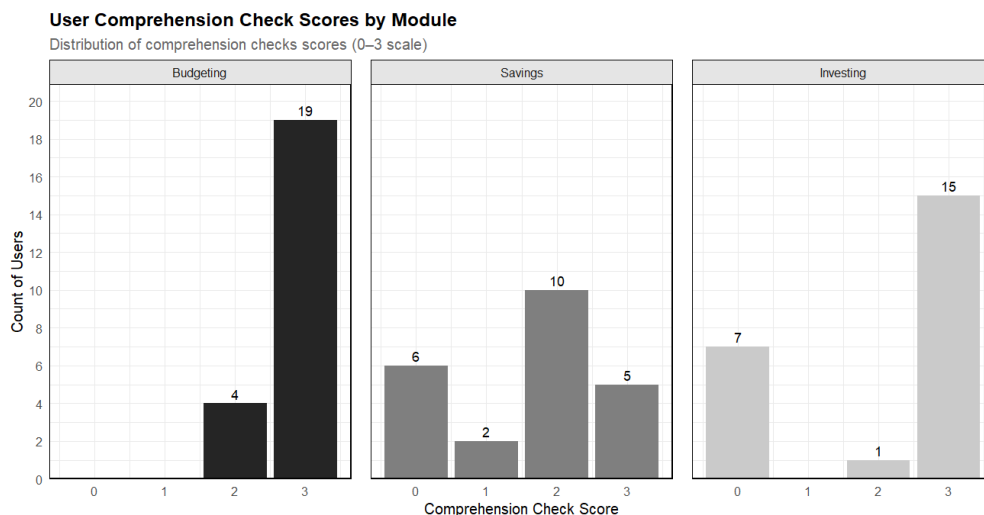


Figure 22: Comprehension check scores for each module

In figure 22 we analyze the comprehension check performance of users in each of the modules with no distinction of which version they received. Each module had 3 comprehension check questions allowing for users to score between 0-3 (See Appendix A2). Both Budgeting and Investing saw most of the users get all three comprehension check questions (note 6 users didn't make it to the investing module and thus scored a 0). Users seemed to struggle most with the Savings comprehension check with only five getting all questions right. This suggests that the savings concepts were the hardest for users to understand.

Additionally, this suggests that there isn't much of a relationship between engagement and performance on comprehension checks. Despite Investing having by far the lowest engagement, users were still able to perform adequately on the comprehension check.

5 Discussion

In this section we will discuss in more detail some of our key findings and suggest reasons why certain things happened in our research study and relate that to the current body of research around financial literacy, online educational experiences and AI use in the learning context.

5.1 AI Coach

The overall usage of the AI coach remained low. Only 6 of the 12 users who had the AI coach version used the coach in some capacity and many of those used it only once or minimally just to test it out. The lack of use of the coach suggests that people, specifically young adults aged 18-25 are still adjusting to the AI world and may prefer more familiar or structured learning experiences [42][43][44]. Many users seemed far more engaged with interactive visualizations and simulations over the AI coach. This shows that users prefer interactive and visual experiences to help them learn instead of personalized conversations with an AI coach [45]. This is in spite of the fact that many users in the pre-survey stated they would be open to using the AI. However, several other users noted their lack of familiarity with AI in the pre-survey.

Despite the lack of overall usage of the AI, there still may be a place for it in online learning experiences. Much of the usage came during comprehension checks. After answering a question incorrectly, the AI could help the user work towards the correct answer. In our review of the user tests, we noticed one user with the non-AI coach version quit the study halfway through after being unable to find the answer to a particular comprehension check question (Savings Q2), showing clear frustration. In contrast, we saw two users from the AI coach version struggle on the same question but use the AI coach to help them get to the correct answer and progress through the study.

This presents an interesting use case for AI in online learning experiences. This finding suggests that AI coaches can be used more as a crutch during learning experiences when the user is struggling with a particular concept [46][47]. This is especially important because ideal learning experiences inspire curiosity and don't invoke frustration from users. User frustration may lead to users quitting or being uninterested in learning a topic as we saw in our study [48]. An AI coach can help remedy this issue by being there to help users with any struggles they have throughout their learning experience and promoting curiosity and avoiding frustration.

Another use case of the AI coach during our study was to clarify certain information or explore a certain topic further. In one particular case, we noticed a user had a distinct curiosity for return on investment

and its different types. This also presents another purpose for AI integration in learning experiences. In traditional online learning experiences, people would have to do their own research of a topic if they wanted more information. However, with an AI coach users have the opportunity to explore certain topics of interest further in the experience itself and can even do so in the context of their learning experience [49].

5.2 Learning Outcomes

In our comparison of users' learning outcomes for those with the AI coach version and those without, we saw relatively even improvement between the two groups. This indicates that the AI coach had no impact on improving users' financial literacy and learning outcomes. This re-enforces the idea that young adults may still be more interested in the more traditional online learning experiences and are not yet ready to use AI to boost their learning despite some research showing AI can improve learning outcomes [50][44].

Another interesting finding here is that young adults may simply not know how to fully use an AI agent to improve their learning outcomes. While AI and chatbots are not necessarily a new concept, they have certainly gained popularity and power in recent years [51]. It is possible that people don't know how to use it to their best benefit yet especially in the context of learning [52]. This is further re-enforced by the fact that several users expressed some unfamiliarity with AI coach usage. This highlights a potential need for teaching people how to use AI to improve learning.

Despite our best efforts to encourage usage of our AI coach, half of those who had access to it still decided not to use it. This means that half of our users from the AI coach version had more or less the same experience as the users without the AI coach. This means comparing learning outcomes between the groups is somewhat imperfect. We acknowledge that a more AI driven learning experience compared to a traditional experience could be a better way to test learning outcomes with and without AI. That said, the users' decision not to use the AI still indicates users may still prefer more traditional and interactive learning experiences over AI learning experiences and that there may just be a lack of experience in using AI for learning.

5.3 Engagement

Similarly to learning outcomes, we saw minimal differences in engagement for the non-AI coach users and AI coach users. Engagement scores for the non-AI users were actually higher than the AI coach users. This signals that an AI coach does not improve user engagement in a learning experience which is contrary to previous research findings [36][37].

We were particularly curious if the AI coach would make the experience more personalized and that users may share more personal information about themselves. Prior research shows that people may be willing to share more detailed information with an AI [33][34]. Because of this, we hypothesized that the AI coach might incite users to share more about their financial situations. However, our study found this not to be the case. As none of our users used the AI coach to input their own financial situation or financial interests.

This is a notable area to research as getting people to share more information about themselves and be more comfortable with AI can help create more personalized experiences for users. Additionally, this information would better help banks and other financial institutions better understand their customers [53]. This study demonstrates that an AI coach in this form does not incite the user to share more personal information.

5.4 User Attention and Engagement

User engagement also decreased throughout the learning experience with the investing module (last module) having the lowest engagement scores of all modules. This indicates that user attention and engagement decreased during the learning experience. This highlights a shortness of attention spans for young adults specifically in the financial literacy context. Prior research echoes the sentiment of shorter attention spans among young adults [54][55]. This is in spite of all the interactive pieces to our learning experience. So, learning experiences should be designed with this in mind and try to deliver material in a short and concise manner. Additionally, learning experiences should include interactive and engaging elements that help keep users' attention.

Another reason the investing module may have had the lowest engagement was due to its information overload. The investing module had an additional interactive visualization compared to the prior two modules and all interactive visualizations in this module were particularly information heavy. Some users even mentioned how the investing module almost contained too much information for them to digest and properly learn from. This highlights the need to design an experience that strikes a balance between material and not overloading the users. Research also indicates similar findings of avoiding overloading the users [56][57].

5.5 Implications for Financial Literacy

In this study our analysis of the pre-survey echoed findings of prior research that budgeting, saving, investing and credit were the most important financial topics for young adults to learn about [8][4]. These

findings emphasize the need to develop financial literacy learning experiences around these topics for young adults as they enter adulthood [6]. As noted previously, financial literacy remains low among young adults and has major financial ramifications for the governments [3][5].

In the three financial literacy test questions in our learning experience, our users were correct on those questions over 90 percent of the time, proving our learning experience helped either build financial knowledge, confidence or intuition to answer these questions correctly. Prior research revealed young adults typically struggled with these questions. In particular, research has shown that many people selected “I don’t know” for these questions highlighting a lack of confidence in financial literacy [3][2]. However, of our users, only one of our users selected that option. This indicates that our learning experience helped users gain confidence in their financial knowledge. Furthermore, this supports evidence that learning experiences developed to improve financial literacy can be successful in accomplishing that goal. This idea is further reinforced by both groups increasing their financial literacy self-assessment in the pre and post survey on average. Overall this study serves as some proof that online learning experiences with interactive data visualizations can improve financial literacy, confidence and intuition.

Interestingly, some users reported decreased financial literacy after going through the learning experience. This potential highlights over-confidence in financial literacy for some users. A decreased self-assessment could mean that after going through the learning modules, users realized their financial literacy wasn’t as high as they previously thought. This is in line with some prior research that states that young adults and men in particular, are likely to overrate their financial literacy [2]. This helps explain why some users decreased their financial literacy rating after the learning modules and it is interesting that this phenomena may even exist in young women as well as men who were previously researched to be underconfident in their financial literacy [2].

5.6 Limitations and Future Work

Throughout this study several limitations and opportunities for future work presented themselves. In our study we were only able to get 25 total users for analysis, two of which had to be dropped for incompleteness. This presented a small sample size prohibiting the generalizability of our results and findings. Another contributor to the small sample size was the time intensive nature of our analysis of each user which required two researchers to each individually review each video in full and agree on scoring for them.

These sample size issues could potentially be resolved in future studies by creating a more standardized approach to engagement scoring or leveraging technological tools to aid with user scoring and analysis.

With a greater sample size findings could be more generalized.

In addition, user quality also presented limitations in our study. Several users prematurely ended the learning experience and others did not follow some of our instructions we set. This presented some issues with analysis as far as how to analyze users who did not complete the full experience or those who did not speak out loud which made it hard to judge their engagement. A potential solution for this issue in future studies could be to have a more controlled study environment where users are monitored live.

Future work of this study would be to increase the sample size of users and to standardize engagement and learning outcomes. More users would make results more generalizable and a more standardized approach to measuring engagement and learning could strengthen findings. Additional work could also include running a similar study with a more AI intense learning experience that forces users to engage and learn with the AI. This could better find the true impact on engagement and learning of AI. Finally, general improvements to the site could be made such as improved user interface for mobile, more financial topic modules, and improved accessibility.

6 Conclusion

In conclusion, we noticed that a participant's self-input knowledge rating gained a very slight positive increase with more data visualization interaction. Those who actively engaged with the data visualizations did see an average of +1 knowledge rating between the pre and post survey in comparison to those who had medium to low interaction, which saw little to no change in knowledge ratings. Interestingly, not many participants made substantial use of the AI coach when they had it available. Only half of the participants who had access to the AI coach ended up even sending a message. Of those who did end up using the AI coach, it was mostly for help with difficult comprehension check questions or to clarify the information being provided in the current lesson. This indicates that although the overall coach usage was low, it could be seen as more of a last resort when the user really needs a bit of extra support to move on to the next part of the module. The value of the AI coach was less in continuously driving engagement like we had planned, but more so in the ability to provide on-demand support for those either struggling or looking for further explanations. Even though user engagement tended to trail off throughout the learning experience, we believe that participants did learn from our site, especially if they did not spend much time prior learning similar concepts. If the modules were more readily available as individual lessons, rather than bundled together, it is possible that users could have been more motivated to interact further with the content. These findings suggest that our site can support financial learning, while showing that any future iterations or similar projects should focus on ways to improve long-term user engagement.

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A Appendices

A.1 User Testing Notes Template

A.1.1 MM/DD/YY (X Participants):

A.1.1.1 Participant X:

Bad Things/Recommended Changes (things in **Red** below):

-

Things to keep an eye on/maybe change needed (things in **Yellow** below):

-

Full Breakdown written as I was watching (things in **Green** are good):

- **Comprehension Check Question 4 Answers (Use exact responses):**

- Budgeting:

- * 1:

- * 2:

- Saving:

- * 1:

- * 2:

- Investing:

- * 1:

- * 2:

- **Random things:**

- **Budgeting Lesson:**

- Intro page:

- Budget chart:

- Comprehension check:

- **Savings Lesson:**

- Intro page:

- Savings dial page:

- Risk option page:

- Comprehension check:

- **Investing Lesson:**

- Intro page:

- Stock simulation page:

- Compounding page:

- Comprehension check:

A.2 Comprehension Check Questions

A.2.1 Budgeting Module

1. Question 1 of 4

Imagine two friends, Alex and Jamie. Alex spends nearly 60% of their income on wants, 35% on needs, and saves very little. Jamie follows the 50/30/20 budgeting rule, setting aside a consistent portion for savings each month.

Based on these habits, which person is more financially secure in the long run, and why?

Person	Wants	Needs	Savings
Alex	60%	35%	5%
Jamie	30%	50%	20%

2. Question 2 of 4

Now, think about Alex's spending. They buy new clothes almost every week, dine out frequently, and upgrade gadgets every year while struggling to build an emergency fund.

In your opinion, what category is Alex overspending in the most, and how could reducing this spending help improve their financial security?

3. Question 3 of 4

You moved to a city where the cost of living is 1/10th higher than where you used to live. For the same salary, how will you be able to keep your savings ratio?

- Increasing purchases by 1/10
- Decreasing purchases by 1/10
- Decreasing purchases by 9/10
- I don't know

4. Question 4 of 4

In your own words, describe the budget visualization shown in this module and the information it showed. 1–2 sentences is enough.

How, if at all, would you use this visualization to support financial decision-making? Please explain. 1–2 sentences is enough.

A.2.2 Savings Module

1. Question 1 of 4

Jim, Noel, Tracy, and Molly are using different strategies to build their emergency savings fund. Who is **NOT** going to build an emergency savings fund quickly?

- **Jim: No Look Savings** – Money directly deposited into their savings account without seeing it.
- **Noel: Impulse Purchasing** – Buys stuff without thought because they think they want or need it.
- **Tracy: 30 Day Savings** – Gives themselves 30 days to think over an impulse purchase.
- **Molly: Pay Yourself First** – Budgets aside a certain amount of their income to savings as a non-negotiable.

2. Question 2 of 4

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?

- More than \$102
- Exactly \$102
- Less than \$102
- Do not know

3. Question 3 of 4

Drag each savings option to their respective risk level.

Available Options:

- Index Funds
- Balanced Mutual Funds
- Certificate of Deposit (CD)
- Individual Growth Stocks
- Treasury Bonds
- Cryptocurrency
- **Low Risk:** Drop items here
- **Medium Risk:** Drop items here
- **High Risk:** Drop items here

4. Question 4 of 4

In your own words, describe the long-term savings visualization shown in this module and the information it showed. 1–2 sentences is enough.

How, if at all, would you use this visualization to support financial decision-making? Please explain. 1–2 sentences is enough.

A.2.3 Investing Module

1. Question 1 of 4

Read John and Jane's investment strategies and select who is more likely to be a successful investor.

- **John**
 - Invests all money into a single stock.
 - Engages in frequent trading, buying and selling stocks multiple times a week.
 - Sells stocks immediately when value drops and chases hype stocks that recently rose.
- **Jane**
 - Spreads her investment across a diverse mix of stocks rather than relying on just one.
 - Keeps her money invested for the long term to ride out normal market ups and downs.
 - Stays calm during market dips, monitoring conditions instead of reacting impulsively.

2. Question 2 of 4

Please tell me if this statement is true or false: "Buying a single company's stock usually provides a safer return than a stock mutual fund."

- True
- False
- Do not know

3. Question 3 of 4

Below are four options for investing a total of \$12,000. Which option is most likely to result in the most overall money yield after 30 years?

- **Option A**
 - Leaves it invested for 10 years.
 - Stops investing afterward.
- **Option B**
 - Invests smaller amounts consistently.
 - Stays invested for 30 years.
 - Allows returns to compound over time.
- **Option C**
 - Invests \$12,000 gradually.
 - Frequently buys and sells.
 - Attempts to time the market.
- **Option D**
 - Waits to invest until income is higher.
 - Invests more money later.
 - Has fewer years to compound.

4. Question 4 of 4

In your own words, describe both investing visualizations shown in this module and the information it showed. 1–2 sentences is enough.

How, if at all, would you use this visualization to support financial decision-making? Please explain. 1–2 sentences is enough.

A.3 Pre/Post Survey Questions

A.3.1 Pre-Survey Questions

1. How would you rate your current financial knowledge? (1–10)
2. Which financial topics would you most like to learn more about? (Select up to 3)
 - Budgeting
 - Saving strategies
 - Investing basics
 - Retirement accounts (401k, Roth IRA, etc.)
 - Credit scores and credit cards
 - Managing debt
 - Taxes
 - Emergency funds
 - Compound interest
 - Financial planning after graduation
3. Please give an estimate of time spent learning about personal finances.
4. During this learning experience, you will interact with financial simulations and visualizations. Which format would you prefer?
 - Using my own real financial habits/data
 - Using hypothetical example scenarios
 - A combination of both
 - No preference
5. How familiar or comfortable are you with using an AI coach?

A.3.2 Post-Survey Questions

1. After going through our site, how would you now rate your financial knowledge? (1–10)
2. Which visualization do you believe was the most engaging/helped you learn the most?
3. If you chose to use the AI coach, why? If not, why not?
4. Do you have any questions, comments, critiques, or additional thoughts about our site that could help us in our development?